

Sudip Sharma

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Education

- Morgan State University**, *Master of Science in Advanced Computing* May 2026 | Baltimore, US
Relevant Coursework: Design And Analysis Of Algorithms, Software Engineering, Cloud Computing
- Kathmandu University**, *Bachelor of Science in Computer Science* 2018 – 2023 | Dhulikhel, Nepal
Relevant Coursework: Data Structures and Algorithms, Algorithms and Complexity, Database Management Systems

Professional Experience

- Center for Equitable Artificial Intelligence and Machine Learning Systems**, *Graduate Mentor (Morgan State University)* May 2025 – Aug 2025 | Baltimore, US
- Led a team of undergraduates on multimodal deep learning pipelines combining 1D-CNN, Transformer, and 2D-CNN architectures for ECG classification, boosting overall model accuracy to 90%+
 - Designed a feature extraction framework across three domains: time, frequency, and time-frequency using **Python**, **NumPy**, and **TensorFlow**, which increased ensemble performance by 14%
 - Mentored students on data preprocessing pipelines (denoising, segmentation, augmentation) reducing model training time by 30% through optimized data loaders and GPU utilization
- EB Pearls**, *Software Engineer* Jul 2023 – Aug 2024 | Lalitpur, Nepal
- Designed and deployed secure authentication systems (login, OTP, token-based authentication) with **SwiftUI** and **JWT**, improving login success rates by 40% and user satisfaction by 63%
 - Integrated Google Calendar API to sync events, improving scheduling efficiency and engagement for 1K+ active users
 - Developed high-performance **GraphQL** APIs in **NestJS**, enhancing data retrieval speed by 37% across mobile and admin platforms
 - Collaborated with cross-functional teams to deploy production-ready features, reducing bug reports by 25% through rigorous testing and code reviews
- EB Pearls**, *Software Engineering Intern* Apr 2023 – Jul 2023 | Lalitpur, Nepal
- Developed iOS mobile apps using **SwiftUI** and **UIKit**, implementing scalable design patterns (MVVM) and boosting in-app session length by 18% through intuitive navigation and animations.
 - Built and optimized **RESTful** and **GraphQL** backend services using **Node.js** and **Express**, improving API response time by 22% via load balancing and query optimization.
 - Prototyped and integrated real-time notification systems using **WebSockets**, enhancing user engagement for core product features by 13%

Projects & Leadership

- 2025 National Symposium on Equitable AI**, Baltimore, US
Research Lead & Presenter, Morgan State University
- Designed and implemented a fairness-aware Convolutional Neural Network (CNN) on the HAM10000 skin lesion dataset using **Python**, **TensorFlow**, and Fairlearn, improving melanoma recall from 29% → 54% while mitigating bias across multiple skin tone subgroups
 - Conducted extensive model interpretability analysis using **Grad-CAM** and **SHAP**, enabling clinicians to visually verify high-risk lesion predictions for better trust in AI outputs
- Mastercard Center for Inclusive Growth**, *AUC Data Science Initiative* Atlanta, US
- Secured **3rd place** by analyzing economic inclusion disparities across demographics using **Python**, **Pandas**, and **Scikit-learn** to build predictive models for financial inclusion metrics
 - Visualized actionable insights using **Matplotlib** and **Seaborn**, helping policymakers understand disparities in digital banking and credit access
- IT MEET 2022**, *Non-tech Events Lead* Dhulikhel, Nepal
- Led the planning and execution of 5+ events including Career Fair, Alumni Meet, Blood Donation Drive, Acoustic Night, and Idea Pitching Competitions, boosting student participation by 21%.

Skills

- Swift**
Frameworks: Swift UI, UIKit
- Python**
Libraries: Matplot.lib, Pandas, Sklearn, Tensorflow
- Javascript/Typescript (Node.js)**
Frameworks: NestJS, Express (GraphQL/REST API)
- Cloud Computing**
Amazon Web Service: S3, EC2, SageMaker, VPC